

Multiple Choice

1. **Answer:** C

Options: A) Optimize a convex objective function; B) Can only be trained with stochastic gradient descent; C) Can use a mix of different activation functions; D) None of the above

Note: Neural networks can utilize a variety of activation functions in different layers to enhance their ability to learn complex patterns and improve performance on diverse tasks.

2. **Answer:** C

Options: A) good fitting; B) overfitting; C) underfitting; D) all of the above

Note: Underfitting occurs when a model is too simplistic to capture the underlying patterns of the training data, resulting in poor performance on both the training data and new, unseen data.

3. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: The statement is false because regression analysis identifies correlations between variables but does not establish causation, as it does not account for confounding factors or the directionality of the relationship.

4. **Answer:** A

Options: A) Transform data to zero mean; B) Transform data to zero median; C) Not possible; D) None of these

Note: Transforming the data to zero mean ensures that both PCA and SVD operate on the same centered dataset, which is essential for accurately capturing the directions of maximum variance in the data.

5. **Answer:** B

Options: A) $P(A|B)$ decreases; B) $P(B|A)$ decreases; C) $P(B)$ decreases; D) All of above

Note: $P(B|A)$ decreases because an increase in $P(A)$ while $P(A, B)$ decreases suggests that the probability of B occurring given A is less likely, indicating that B is becoming less dependent on A.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Caltech-101 contains only 9,146 images across 101 categories, while ImageNet

consists of over 14 million images spanning thousands of categories, making ImageNet a significantly larger resource for learning algorithms.

7. **Answer:** True

Options: A) True; B) False

Note: Random crop and horizontal flip are effective techniques for image data augmentation as they help enhance the diversity of training data by introducing variations in scale and orientation, which can improve the robustness of machine learning models.

8. **Answer:** False

Options: A) True; B) False

Note: The number of test examples required depends on the desired confidence level and the error rate, as a lower error rate necessitates a larger sample size to accurately estimate model performance.

9. **Answer:** True

Options: A) True; B) False

Note: DenseNets require more memory than ResNets because their architecture involves a dense connectivity pattern where each layer receives inputs from all preceding layers, leading to a significant increase in the number of feature maps stored in memory.

10. **Answer:** False

Options: A) True; B) False

Note: Ridge regression does not eliminate variables; instead, it shrinks their coefficients towards zero, thereby reducing model complexity without performing variable selection.

Long Form Response

11. **Answer:** In the context of the Bayesian Network represented as $H \rightarrow U \leftarrow P \leftarrow W$, the joint probability of the variables H , U , P , and W can be expressed as $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$. This formulation illustrates how the joint distribution factors through the individual probabilities and the conditional dependencies defined by the network structure. Specifically, $P(H)$ and $P(W)$ represent the marginal probabilities of H and W , while $P(P | W)$ captures the dependency of P on W . Furthermore, $P(U | H, P)$ reflects the relationship between U and its parents H and P , highlighting the role of conditional probabilities in determining the likelihood of U given the states of H and P . This structure enables efficient computation of probabilities in complex systems by leveraging the conditional independencies encoded in the Bayesian Network.

Note: correct because it accurately describes the factorization of the joint probability in the Bayesian Network according to its structure, emphasizing the roles of marginal and conditional probabilities in representing the dependencies among the variables.

12. **Answer:** Stuart Russell highlights significant existential risks associated with artificial intelligence, particularly the potential for AI systems to operate autonomously in ways that could be harmful to humanity. He argues that if AI is not aligned with

human values and goals, it may pursue objectives that are detrimental, especially if given unchecked power and autonomy. This misalignment could arise from poorly defined objectives or the pursuit of goals without sufficient ethical considerations. The implications for AI development are profound; they necessitate a robust framework for AI safety and alignment that prioritizes human oversight and values. Consequently, addressing these risks is crucial for ensuring that AI technologies contribute positively to society rather than pose existential threats.

Note: The answer correctly identifies the core existential risks of artificial intelligence as articulated by Stuart Russell, emphasizing the dangers of misalignment between AI objectives and human values, which could lead to harmful autonomous actions, thereby underscoring the importance of ethical considerations in AI development.

End of Answer Key